Interim Progress Report 02

EECS 4312 Project: Requirement Verification for Smart Health & Wellness Center Automation

Date: July 25, 2025
Team Members: Kanwarjot Singh Bharaj (214284269), Bhavneet Kaur (218501361), Jaideep Singh (218472241)

1. Work Completed So Far

Task 2: LTL Formulation and SPIN-Based Model Checking

- **Jaideep**:
    - Supported property translation from English to formal LTL syntax for both scheduling and payment scenarios.
    - Assisted in refining system logic abstraction into PROMELA code blocks.
    - Participated in reviewing the accuracy of SPIN model behavior during execution cycles.
- **Bhavneet**:
    - Authored LTL properties for scheduling subsystems, including double booking prevention and slot modification safety.
    - Developed a PROMELA model for scheduling logic, simulating booking requests and conflict resolution.
    - Executed SPIN verification commands and documented outputs.
    - Key Contribution:
        - Created property LTL1: [](booked → !booked_by_others)
        - Created property LTL4: [](modify → (!booked_by_others))
- **Kanwarjot**:
    - Focused on the payment and billing module.
    - Created LTL properties to ensure payment confirmation precedes access, and insurance verification occurs before booking.
    - Developed and tested a PROMELA model for payment workflows.
    - Automated SPIN runs using shell scripting to verify property satisfaction and check for runtime violations.
    - Key Contribution:
        - Created property LTL2: [](access → paid)
        - Created property LTL3: [](booking_requested ∧ !insurance_verified → X !booking_confirmed)

2.Current Responsibilities

Team Members	Tasks	Deliverables
Jaideep	Draft Section 1 & 2 of the final report (overview, requirement extraction analysis). Document assumption justifications and stakeholder role mapping. Help compile and organize code/scripts into appendix.	Final report – Sections 1 & 2. Stakeholder table & assumption rationale. Appendix: NLP scripts and requirement tables.
Bhavneet	Finalize and review LTL properties for scheduling. Rerun SPIN models with multiple user scenarios and document results. Draft Section 3 of the final report (SPIN & LTL for scheduling).	Final report – Section 3 (scheduling module). SPIN screenshots and logs. Updated scheduling LTL list.
Kanwarjot	Review and validate SPIN model for payment edge cases (e.g., partial payments, late verification). Draft Section 4 (analysis of model checking results). Assist in formatting and merging full reports.	Final report – Section 4 (conflict analysis). SPIN output logs (payment module). Refined billing LTL properties.

3. Plan for Next Week (July 25 - 31)

Jaideep

- Draft and finalize Section 1 (Project Overview) and Section 2 (Requirement Extraction & Analysis) by July 27.
- Document stakeholder roles and assumptions in tabular format for the final report appendix.
- Collaborate with Bhavneet to compile all verified requirement artifacts and NLP scripts.

Bhavneet

- Rerun SPIN simulations for scheduling module with multiple users and edge scenarios.
- Add runtime assertions (`assert()`) to strengthen SPIN validation coverage.
- Draft Section 3 of the final report (SPIN + LTL Modeling for Scheduling).
- Compile screenshots and logs from SPIN runs by July 29.

Kanwarjot

- Refine LTL properties and test logic for billing exceptions (e.g., “service access without invoice”, “delayed insurance approval”).
- Finalize PROMELA model and automate test cases for the billing module.
- Draft **Section 4** of the final report (Model Checking Analysis + Conflict Resolution).
- Assist in final report formatting and appendix compilation.

Team-Level Goals

- Merge individual report sections into a cohesive document by July 30.
- Add final appendix: LTL files, PROMELA models, SPIN screenshots, NLP outputs.
- Submit the completed project report + ZIP file to eClass by **Aug 01.**